Beyond Surveys: Predicting Student Well being from Open ended Video Responses using LLMs

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Abstract

Traditional mental health assessment relies on infrequent self-report questionnaires (e.g., PHQ-9), which provide only sparse, retrospective snapshots of well-being and may miss everyday fluctuations. This creates an opportunity for low-burden, lightweight assessments embedded in everyday interactions, such as open-ended conversations with AI systems, to complement existing approaches and enable more natural, potentially more reliable monitoring of well-being over time. However, it remains unclear whether large language models (LLMs) can predict self-reported psychological well-being from naturalistic multimodal data. As a first step toward such an AI-based informal assessment system, we recruited N=228 college students to collect self-reported well-being responses paired with corresponding open-ended video responses. Using this dataset, we conducted four studies to examine the effects of input modalities, model architectures, contextual inputs, and prompting strategies. Our results show that LLMs leveraging multimodal features outperform regression baselines without task-specific fine-tuning, and that retrieval-augmented chain-of-thought prompting with contextual inputs yields the most robust performance. These findings show the promise of multimodal LLM-based approaches for scalable, low-burden well-being assessment and offer design guidance for integrating conversational AI into real-world affective computing systems.

CCS Concepts

• **Human-centered computing** → **User models**; • **Applied computing** → *Health care information systems*; • **Computing methodologies** → *Natural language generation.*

Keywords

Conversational Agent, LLMs, Mental Health, Well-being Assessment, Multimodal Analysis, Prompt Engineering, Affective Computing

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1 Introduction

Mental disorders remain a leading cause of global disability, imposing substantial costs on individuals, healthcare systems, and society at large [79]. Standard mental health and psychological assessments rely heavily on self-report questionnaires, such as PHQ-9 [33] and GAD-7 [65], which provide cost-effective and standardized guidelines for diagnosis and treatment planning. However, these instruments are often limited by retrospective recall bias and are poorly suited to capture the variability of mental health symptoms and the contextual factors that shape everyday well-being [48]. These limitations underscore the need for low-burden, scalable methods for assessing psychological well-being, a challenge that is both a pressing public health imperative and a fundamental scientific problem [52, 61]. One prominent response to these challenges has been the use of ecological momentary assessment (EMA) [62], which involves repeatedly sampling individuals' psychological states in their natural environments to capture in-situ experiences. However, EMA remains reliant on repetitive self-report and can introduce burden and disengagement over time [51].

Although prior work [46] suggests that integrating multiple modalities, such as text, audio, and visual cues, can improve detection of affective and mental health conditions over unimodal approaches by leveraging complementary behavioral information, traditional machine learning models often struggle to effectively represent and fuse these heterogeneous, high-dimensional signals, especially when the input is spontaneous, open-ended, and weakly structured [16, 46, 60, 67]. Although deep learning improves multimodal modeling [55, 58, 89], it has critical bottlenecks: heavy reliance on scarce, private clinical data [12], lack of interpretability, and narrow task-specificity [82]. These challenges are important in mental health, where datasets are often small, sensitive, and imbalanced [46], making the end-to-end training of deep learning models from scratch difficult and increasing the risk of poor generalization or out-of-distribution failure [25]. Recently, pre-trained Large Language Models (LLMs) have shown promise for text-based mental health prediction, primarily targeting acute clinical risk, such as depression or suicide ideation, through task-specific fine-tuning or specialized prompting [26, 64, 72, 80]. Yet, their potential for broader well-being assessment from naturalistic, multimodal behavior remains underexplored [20, 22, 28]. Thus, we investigate whether *LLMs can support low-burden well-being prediction from multimodal inputs* derived from self-recorded videos of open-ended responses. Specifically, we address the following research questions:

- **RQ1**. Does integrating visual and acoustic non-verbal cues improve LLM-based well-being assessment compared to text-only input?

- **RQ2**. How do LLM-based approaches compare to other machine learning techniques previously used in psychological assessment prediction?
- **RQ3**. Does incorporating contextual information from overall well-being-related video responses improve LLM-based prediction compared to using only a single survey-specific video response?
- **RQ4**. How do various prompting strategies and techniques improve the LLM's performance in well-being assessment prediction tasks?

2 Related Work

2.1 Digital Mental Health Assessment and Multimodal Behavioral Signals

Mental health and well-being are increasingly understood as dynamic, context-dependent phenomena that are difficult to capture through static, retrospective self-reports alone [69, 74, 77]. Ecological Momentary Assessment (EMA) [62] has emerged as a more ecologically valid alternative by enabling in-situ measurement of moment-to-moment experiences and has been particularly useful for studying fluctuating constructs, such as stress and well-being [5, 43, 77]. However, these approaches remain dependent on repeated self-report and are often constrained by declining user engagement caused by repetitive prompts, survey fatigue, and the limited expressiveness of structured responses [51]. These limitations motivate low-burden approaches that can infer well-being from naturally expressed behavior. Thus, in this work, we explore the potential efficacy of open-ended responses, in which people can more naturally express themselves, in predicting their well-being states that are typically assessed in structured, closed-ended self-reported surveys.

In parallel, psychological well-being is expressed through multiple behavioral channels, including language, vocal, and facial behavior [32, 54, 76]. Prior work shows that mental and affective states are reflected in nonverbal behavior [54, 76] and speech-related patterns [32], motivating multimodal approaches that integrate verbal and non-verbal signals more robustly than unimodal systems alone [14, 46, 54, 57, 67]. Building on this, our work tests whether multimodal signals extracted from open-ended video responses can support LLM prediction of validated well-being surveys.

2.2 AI/ML-based Mental Health Prediction

Mental states are inherently complex and expressed across multiple modalities [32, 54, 76]. Thus, Artificial Intelligence and Machine Learning (AI/ML) approaches to mental health have evolved from unimodal analyses of isolated signals, such as acoustic markers of depression, lexical indicators of distress, or reduced facial expressivity, to multimodal frameworks that integrate linguistic, acoustic, visual, and behavioral cues for more robust inference [1, 14, 54, 57, 67, 81, 83, 90]. Traditional multimodal systems have largely relied on foundational signal processing techniques [6] and regression-based models, which often struggle with the non-linear, high-dimensional nature of psychological data and unstructured inputs such as spontaneous speech [16, 60]. Recent advancements have driven the field toward deep learning architectures to model the latent relationships within multimodal data [55, 58, 89]. Despite their success, training these networks from scratch introduces critical bottlenecks. They require extensive, expertly annotated clinical datasets, which are scarce due to severe privacy restrictions [12]. Moreover, standard deep learning models suffer from a profound lack of algorithmic transparency, failing to provide the crucial interpretability required for sensitive psychological contexts, and remain largely confined to narrow, task-specific applications [82]. Large Language Models (LLMs) and advanced Natural Language Processing (NLP) techniques have recently shown strong performance in mental health prediction from text [29, 40] and, more recently, multimodal inputs [28]. Comprehensive systematic reviews emphasize that the future of emotion-aware conversational AI relies heavily on integrating text, speech, and behavioral signals [39]. Recent computational advances have begun to operationalize this fusion directly within LLM architectures. For example, studies comparing audio and text modalities for depression and PTSD detection found that combining GPT-4o [49] with acoustic baselines yields superior F1 scores compared to unimodal text analysis [2]. Furthermore, recent frameworks such as [88] improve depression screening by integrating visual encoders into audio-language models beyond the text-centric limits of standard LLMs.

Although current work primarily targets acute clinical pathologies, such as depression and suicidal ideation [72, 80], and focuses on using clinical interviews [11, 86], comparatively little attention has been given to broader well-being assessment from naturalistic, multimodal behavior. To address this gap, our work evaluates untuned LLMs for multimodal well-being assessment from open-ended video responses. By systematically examining modality integration, model architecture, and prompting strategies, we investigate the potential of LLM-based conversational agents as scalable, low-burden tools for high-frequency, ecologically valid mental health monitoring beyond acute clinical risk.

2.3 Prompting and Representing Multimodal Evidence for LLMs

Beyond multimodal feature integration, LLM performance in both general language tasks [10] and psychometric assessment depends strongly on prompting strategy [30, 63, 82]. [30] proposes a modular prompt design for emotion and mental health tasks including persona, task instruction, n-shot examples, input, output, and template, and shows that these choices can substantially affect performance. We later build on our prompt based on its guidance. Chain-of-Thought (CoT) prompting elicits intermediate reasoning steps [31, 78], which is critical for interpretability in mental health contexts [82]. Retrieval-Augmented Generation (RAG) enhances LLM reasoning by grounding predictions in relevant reference cases or clinical benchmarks, improving calibration and reducing hallucinations [35, 63]. Thus, we evaluate each strategy separately, as well as their combination. Motivated by combining integrates reasoning with empirical grounding frameworks (RAG+CoT) [71, 75], we examine if it can yield more reliable predictions across diverse and skewed well-being measures.

3 Participants and Data Collection

We recruited N=228 participants from the [Anonymous] University psychology subject pool (159 M, 67 F, 1 Other, 1 Prefer not to say; age M=19.05, SD=1.75, $range$=18–40). The sample was racially diverse (49.1% White/Caucasian, 35.1% Asian, 2.6% Black/African-American, and 13.2% Multiracial, Hispanic, or Other), mostly native English speakers (79.8%), and drawn from a range of undergraduate majors, including STEM, business, and health-related fields. The study was conducted entirely online via Qualtrics (qualtrics.com).

Once the consent was electronically signed, participants completed nine validated self-report assessments regarding their overall mental health and well-being: Generalized Anxiety Disorder-7 (GAD-7) [65], Meaning in Life Questionnaire (MLQ) [66], Multidimensional Scale of Perceived Social Support (MSPSS) [91], Patient Health Questionnaire-9 (PHQ-9) [33], Perceived Stress Scale (PSS) [13], Scale of Positive and Negative Experience (SPANE) [19], Sleep Quality Scale (SQS) [84], Satisfaction With Life Scale (SWLS) [18], and the UCLA Loneliness Scale (UCLA) [59]. These self-report responses served as ground-truth labels for the well-being prediction task for our study.

Next, participants were asked to record themselves responding to nine open-ended questions, each corresponding to one of the well-being surveys described above. The video-response questions were designed to be open-ended and suitable for conversational elicitation (e.g., the question corresponding to GAD-7: "Over the past two weeks, what are some anxious or worried thoughts you've had? How often have these thoughts or feelings affected you?"). Our study was approved by the [anonymous] University Institutional Review Board. Participants received either 1.5 course credits or a $10 Amazon gift card after passing internal attention and screening checks. Additional recruitment and demographic details are in Section A. Section D.2 lists all video-response prompts, and Section B describes the train/-validation/test split used across all methods and baselines.

4 Multimodal Feature Extraction

Since current LLMs operate primarily over textual inputs, we translate nonverbal video signals into compact, human-readable information that can be integrated directly into the prompt. For each participant, all nine open-ended video responses were processed into three distinct signal streams (Figure 1): **(1) transcripts**, generated using Microsoft Azure AI Speech services [45]; **(2) visual cues**, consisting of facial blendshapes (e.g., eye blinks, jaw openness, and specific mouth movements) extracted using Google's MediaPipe [37]; and **(3) acoustic cues**, also extracted via Azure AI Speech services [45], comprising speech assessment metrics (e.g., prosody, fluency, pronunciation, and completeness scores). To capture the most salient facial expressions while minimizing noise, the visual features were filtered to retain only the top five highest-intensity blendshapes per frame. Finally, both the visual and acoustic cues were summarized using standard descriptive statistics (count, mean, standard deviation, min/max, quartiles) and combined with the transcripts to form a unified, text-compatible multimodal input.

5 Evaluation Metrics

We evaluate model performance against ground-truth survey scores using three complementary metrics. **Category accuracy** measures how accurately models classify participants into standard clinical severity bins derived from predicted scores. We also compute **Matthews Correlation Coefficient (MCC)** [42] as a balanced classification metric robust to class imbalance [17], and **Pearson correlation** ( ) [53] to assess linear association between predicted and ground-truth continuous scores. In the four main study sections (Sections 6 to 9), we report category accuracy results. Pearson correlation and MCC results are discussed in Section 10. For categorical accuracy, we include a RandGuess baseline (defined as $1/C$ for each survey), which is excluded from Pearson and MCC analyses because random predictions provide no meaningful correlation or discrimination.

6 Study 1: Effects of Multimodal Cues on LLM-Based Well-Being Assessment

In this first study, we address **RQ1:** *"Does integrating visual and acoustic non-verbal cues improve LLM-based well-being assessment in comparison to transcript-only input?"*. To isolate the effect of modality, we use a fixed LLM (GPT-5.2 [50]) and zero-shot prompt while varying only the input modality configuration from unimodal to multimodal over the same participant data.

Method. We compare a unimodal, transcript-only input against multimodal variants that combine the same transcripts with summarized facial cues, summarized acoustic cues, or both. Prior work [80] has shown that transcript-only language features can support mental health prediction, motivating our transcript-only baseline. The unimodal setting (GPT-T0) uses *only* transcripts from the nine videos. The multimodal variants add *only* visual cues (GPT-TF0), *only* acoustic cues (GPT-TA0), or *both* (GPT-TFA0). For each survey, a prediction is made using the single video response corresponding to self-reported survey responses; for example, GAD-7 is predicted from the participant's response to the GAD-7 question. All evaluated input configurations use a fixed zero-shot prompt. Following [30], the prompt generally includes an agent persona (a clinical psychologist evaluating a participant's well-being), a task description (the target survey), and step-by-step instructions (analyze verbal responses together with facial and acoustic cues, then provide a score, interpretation, and confidence level). The full prompt format and an example are provided in Section F.

Results. Table 1 shows that LLM prediction with all different input configurations outperforms the random baseline (RandGuess) on all surveys except the SQS. This indicates that naturalistic, open-ended video responses contain meaningful psychometric signals that LLMs can effectively interpret. Transcript-only input (GPT-T0) establishes a strong baseline, suggesting that explicit verbal disclosures capture the majority of the variance needed for psychological assessment. This suggests that transcript-only input can serve as a strong baseline, since LLMs are particularly effective at extracting patterns from language when participants express the relevant signal directly in words [4, 23].

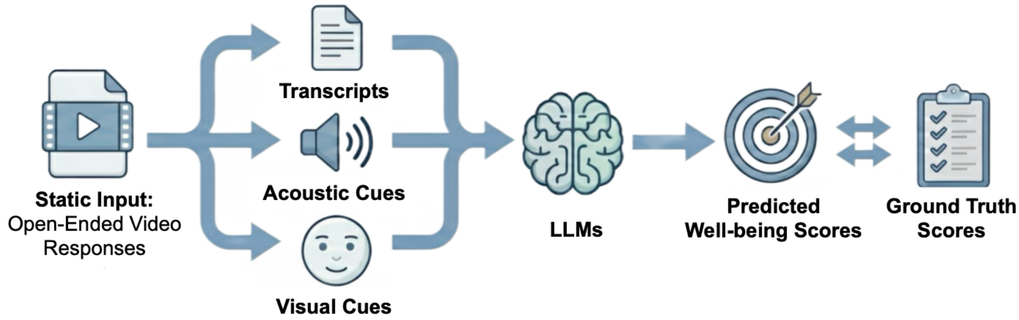

Figure 1: Offline multimodal assessment pipeline. Each participant provided seven open-ended video responses, from which transcripts, acoustic cues, and visual cues are extracted and summarized as input to LLMs for predicting validated well-being survey scores.

Table 1: [Study 1] Category prediction accuracy (mean ± std over three seeds) across input configurations. Best values per column are in bold. Full multimodal input performs best overall, though transcript-only and partially multimodal variants remain competitive on some surveys. GPT-T0 refers to *text-only* unimodal setting. GPT-TF0 refers to the *text visual* multimodal variants with visual cues GPT-TA0 refers to the *text audio* variants. GPT-TFA0 refers to the *text visual acoustic* variants.

Method	GAD-7	MLQ	MSPSS	PHQ-9	PSS	SPANE	SQS	SWLS	UCLA
RandGuess	0.25	0.20	0.33	0.20	0.33	0.20	**0.20**	0.14	0.25
GPT-T0	0.35 ± 0.01	0.58 ± 0.02	0.77 ± 0.01	0.59 ± 0.01	**.63 ± .**	0.41 ± 0.01	0.10 ± 0.01	0.30 ± 0.02	0.40 ± 0.01
GPT-TF0	0.35 ± 0.00	**.61 ± .**	0.77 ± 0.01	0.57 ± 0.01	**.63 ± .**	0.41 ± 0.02	0.13 ± 0.03	0.32 ± 0.01	0.41 ± 0.01
GPT-TA0	0.36 ± 0.01	0.60 ± 0.01	0.77 ± 0.00	**.6 ± . 1**	**.63 ± .**	0.43 ± 0.01	0.14 ± 0.02	0.31 ± 0.01	0.41 ± 0.00
GPT-TFA0	**.37 ± . 1**	**.61 ± . 1**	**.78 ± . 1**	**.6 ± . 1**	**.63 ± . 1**	**.44 ± . 1**	0.13 ± 0.04	**.32 ± . 1**	**.42 ± . 2**

Integrating nonverbal information yields consistent, complementary improvements, underscoring the value of multimodal input. Adding visual cues alone (GPT-TF0) improves performance on MLQ (+0.03), SWLS (+0.02), and UCLA (+0.01) compared to GPT-T0. This suggests that facial cues, such as smile intensity or reduced expressivity, reveals subtle indicators of life satisfaction, meaning, and social-emotional state that are not fully captured in transcripts alone [7]. Besides, adding acoustic cues alone (GPT-TA0) improves or preserves performance across nearly all surveys relative to the transcript-only baseline, producing the strongest partially multimodal results on PHQ-9 (0.60), SPANE (0.43). In particular, compared with GPT-T0, acoustic cues yield gains on GAD-7 (+0.01), SPANE (+0.02), and SQS (+0.04), while preserving the strongest PSS performance at 0.63. This pattern indicates that vocal delivery carries complementary information of distress and affective state, aligning with clinical literature identifying vocal prosody, speech rate, and pause duration as robust biomarkers for depression [44, 87].

The full multimodal configuration (GPT-TFA0) performs best overall, achieving the highest or tied-for-highest accuracy on eight of the nine surveys. Although the marginal gains partially multimodal settings are sometimes modest (typically +0.01 to +0.03), the improvement across diverse psychological constructs suggests that the three modalities likely contribute complementary, rather than redundant signals. Taken together, predicting well-being benefits significantly from jointly modeling *what* participants say alongside *how* they express themselves audibly and visually.

However, the Sleep Quality Scale (SQS) remains challenging, with performance ranging only from 0.10 to 0.14 across LLM configurations, compared with RandGuess at 0.20. This suggests that sleep-related constructs may be less readily captured from brief open-ended video responses in the current setting, or that the cues most relevant to sleep quality are expressed less directly in spontaneous verbal and nonverbal behavior than the other well-being measures considered here. It may also indicate that zero-shot prompting with untuned LLMs is not yet sufficient for SQS and that more advanced inference strategies are needed.

7 Study 2: Model Architecture Comparison

Results from Study 1 (Section 6) showed that incorporating multimodal inputs improves LLM-based well-being assessment. Thus, Study 2 aims to address **RQ2:** *"How do LLM-based approaches compare to other machine learning techniques previously used in psychological assessment?"*. To answer this question, we compare the best-performing multimodal LLM from Study 1 with supervised regression models and VLMs under a shared multimodal input setup, isolating differences due to model architecture and inference rather than input content or data availability.

Method. We benchmark GPT-TFA0 against two regression baselines, two VLMs, and two alternative LLMs. The **regression baselines**, Ridge [27] (Ridge) and Random Forest [9] (RandForest), use LIWC-22 [8] transcript features combined with the same acoustic and facial features as

Table 2: [Study 2] Category prediction accuracy (mean ± std over three seeds) across all models in three studies. Best values in each column are shown in bold. LLM-based models with multimodal inputs without task-specific fine-tuning perform best overall, though regression remains competitive on some surveys.

	Method	GAD-7	MLQ	MSPSS	PHQ-9	PSS	SPANE	SQS	SWLS	UCLA
	RandGuess	0.25	0.20	0.33	0.20	0.33	0.20	0.20	0.14	0.25
Regres.	Ridge	**.4 ± .**	0.52 ± 0.00	0.63 ± 0.00	0.37 ± 0.00	0.42 ± 0.00	0.31 ± 0.00	0.23 ± 0.00	0.17 ± 0.00	0.27 ± 0.00
	RandForest	0.37 ± 0.02	0.57 ± 0.02	0.74 ± 0.00	0.28 ± 0.01	**.65 ± .**	0.46 ± 0.01	**.41 ± . 1**	0.24 ± 0.02	0.42 ± 0.01
VLMs	LlaVA	0.15 ± 0.03	0.54 ± 0.08	0.43 ± 0.12	0.11 ± 0.01	0.54 ± 0.02	0.24 ± 0.04	0.08 ± 0.02	0.17 ± 0.07	0.20 ± 0.03
	Qwen	0.22 ± 0.03	0.55 ± 0.01	0.60 ± 0.03	0.27 ± 0.02	0.48 ± 0.03	0.38 ± 0.02	0.11 ± 0.04	0.17 ± 0.02	0.30 ± 0.02
LLMs	DeepSeek	0.37 ± 0.11	0.56 ± 0.03	0.76 ± 0.05	0.56 ± 0.06	0.49 ± 0.03	0.41 ± 0.16	0.11 ± 0.03	0.32 ± 0.03	**.48 ± . 3**
	Llama	0.22 ± 0.01	0.59 ± 0.00	0.76 ± 0.00	0.47 ± 0.01	0.58 ± 0.02	**.47 ± . 1**	0.14 ± 0.00	**.35 ± .**	0.45 ± 0.01
	GPT-TFA0	0.37 ± 0.01	**.61 ± . 1**	**.78 ± . 1**	**.6 ± . 1**	0.63 ± 0.01	0.44 ± 0.01	0.13 ± 0.04	0.32 ± 0.01	0.42 ± 0.02

GPT-TFA0. These baselines are motivated by prior work using structured linguistic and behavioral features for mental health inference [24, 47]. We also evaluate two **VLMs**, Qwen3-VL:32B [3, 68, 73] (Qwen) and LLaVA1.6:7B [36] (LLaVA), which process raw video frames, rather than summarized facial cues, together with transcripts and acoustic cues. We include VLMs because recent works demonstrate their superior capacity for spatiotemporal reasoning and direct video understanding compared to text-only LLMs [38, 85]. To assess generalization across LLM architectures, we evaluate two additional **LLMs**: DeepSeek-R1:70B [15] (DeepSeek) and LLaMA-4-Maverick [70] (Llama). All LLM- and VLM-based models use the same zero-shot prompt as in Section 6. Across all model families, prediction for each survey is based on the single video response corresponding to that survey, consistent with the setup in Section 6.

Results. Table 2 shows that all evaluated models outperform RandGuess across all surveys, indicating that naturalistic video responses contain meaningful psychometric signals. Overall, LLM-based models with multimodal inputs and no task-specific fine-tuning perform best.

At the same time, the regression baselines remain competitive on selected measures. Ridge performs best on GAD-7 (0.40), while Random Forest performs strongly on PSS (0.65). Most notably, both regression models outperform RandGuess on SQS, and Random Forest achieves the best SQS performance overall at 0.41, compared with only 0.13-0.14 for the strongest LLMs. This suggests that sleep-related conditions may still be partially captured from open-ended video responses, addressing the challenge raised in Section 6, although the present study does not isolate which specific cues drive that performance. At the same time, it indicates that zero-shot prompting with untuned LLMs is not yet sufficient for SQS, and that supervised regression models may capture or use these cues more reliably. Beyond this exception, however, LLMs remain stronger overall.

The VLM results further reinforce this pattern. Although Qwen outperforms RandGuess across all surveys and generally performs better than LlaVA, both VLMs remain weaker than the strongest LLM-based approaches on most constructs. In particular, LlaVA performs poorly on several distress-oriented measures such as GAD-7 (0.15) and PHQ-9 (0.11), while Qwen shows only moderate gains, reaching 0.22 on GAD-7 and 0.27 on PHQ-9. Compared with both regression baselines and VLMs, all three LLMs show stronger and more consistent performance across most well-being constructs, suggesting that multimodal prompting provides a more robust general-purpose approach for this task.

Among the tested models, tuning-free GPT with multimodal inputs (GPT-TFA0) performs best across all model classes, achieving the highest or tied-highest category accuracy on MLQ (0.61), MSPSS (0.78), and PHQ-9 (0.60). Its gains over the other LLMs are modest but consistent, for example, it improves over Llama on MLQ (0.61 vs. 0.59), MSPSS (0.78 vs. 0.76), and PHQ-9 (0.60 vs. 0.47), while remaining competitive with DeepSeek and Llama on the remaining surveys. Taken together, these findings demonstrate that neither direct VLM-based video processing nor traditional regression is optimal for comprehensive well-being assessment. Instead, the strongest performance is achieved by explicitly extracting and combining multimodal signals from videos with the reasoning capabilities of LLMs.

8 Study 3: Single Survey-Specific Video vs. All-Video Input Comparison

Having shown that LLMs with multimodal feature representations outperform other model architectures, we next test whether aggregating multiple open-ended video responses improves performance over using only the target survey response, as in the previous two studies (Sections 6 and 7), motivated by psychometric evidence that stable psychological traits are more reliably inferred from behavior aggregated across situations than from a single observation [21]. Accordingly, we address **RQ3:** *"Does aggregating multimodal information across all video responses improve LLM-based well-being prediction compared to using only a single survey-specific response?"*. To isolate the effect of input context aggregation, we use a fixed LLM, GPT-5.2 [50], together with the fixed zero-shot prompt used in Sections 6 and 7.

Method. We compare two input settings using the same tuning-free GPT-based multimodal prompting approach with a fixed zero-shot prompt for each survey. In the *one-video* setting (GPT-TFA0), each survey score is predicted from the single video response corresponding to that survey prompt, consistent with the setup used in the previous two studies (Sections 6 and 7). For example, GAD-7 prediction is based only on the participant's response to the GAD-7 open-ended question. In the *all-videos* input setting (GPT-TFA0-All), the model predicts each survey score using

Table 3: [Study 3] Category prediction accuracy (mean ± std over three seeds) for tuning-free GPT with multimodal inputs using either the survey-specific *one-video* (GPT-TFA0) input or *all-video* (GPT-TFA0-All) aggregated input. Best values per column are shown in bold. Aggregating behavior across all video responses improves performance on most surveys.

Method	GAD-7	MLQ	MSPSS	PHQ-9	PSS	SPANE	SQS	SWLS	UCLA
RandGuess	0.25	0.20	0.33	0.20	0.33	0.20	0.20	0.14	0.25
GPT-TFA0	0.37 ± 0.01	0.61 ± 0.01	0.78 ± 0.01	**.6 ± . 1**	0.63 ± 0.01	0.44 ± 0.01	0.13 ± 0.04	0.32 ± 0.01	0.42 ± 0.02
GPT-TFA0-All	**.38 ± . 1**	**.66 ± .**	**.81 ± . 1**	**.6 ± . 1**	**.71 ± . 1**	**.57 ± . 1**	**.15 ± . 4**	**.37 ± .**	**.67 ± . 1**

Table 4: [Study 4] Category prediction accuracy (mean ± std over three seeds) across LLM prompting strategies. Best values per column are shown in bold. Advanced prompting improves performance over zero-shot prompting, with RAG+CoT performing best overall.

Method	GAD-7	MLQ	MSPSS	PHQ-9	PSS	SPANE	SQS	SWLS	UCLA
RandGuess	0.25	0.20	0.33	0.20	0.33	0.20	0.20	0.14	0.25
GPT-TFA0-All	0.38 ± 0.01	0.66 ± 0.00	0.81 ± 0.01	0.60 ± 0.01	0.71 ± 0.01	0.57 ± 0.01	0.15 ± 0.04	0.37 ± 0.00	0.67 ± 0.01
GPT-CoT-All	0.50 ± 0.02	0.60 ± 0.01	0.74 ± 0.01	0.56 ± 0.02	0.72 ± 0.01	0.59 ± 0.05	0.27 ± 0.03	0.43 ± 0.02	0.61 ± 0.01
GPT-Fewshot-All	0.41 ± 0.01	0.61 ± 0.00	0.80 ± 0.00	0.59 ± 0.01	**.74 ± . 1**	0.59 ± 0.03	0.44 ± 0.01	0.35 ± 0.01	**.68 ± . 1**
GPT-RAG-All	0.47 ± 0.01	0.64 ± 0.02	0.80 ± 0.00	**.61 ± . 1**	0.70 ± 0.02	0.58 ± 0.02	0.42 ± 0.02	0.39 ± 0.02	0.67 ± 0.01
GPT-RAGCoT-All	**.54 ± . 1**	**.67 ± . 1**	**.82 ± .**	**.61 ± . 3**	**.74 ± . 1**	**.62 ± . 3**	**.45 ± . 1**	**.45 ± . 1**	0.63 ± 0.01

information aggregated from all nine video responses collected from the participant. For example, GAD-7 prediction is based on the participant's response to the GAD-7 question together with the other eight responses corresponding to the remaining surveys. In both settings, the model receives the same full multimodal input, consisting of transcript content together with summarized facial and acoustic cues.

Results. Table 3 shows that using all video responses improves or matches performance on all surveys, with the largest gains on UCLA (+0.25), SPANE (+0.13), and PSS (+0.08), suggesting that broader behavioral context yields more reliable well-being estimates than a single survey-matched response and emperically supporting prior psychometric evidence that stable traits are better inferred from behavior aggregated across situations [21]. However, SQS remains challenging: GPT-TFA0-All performance still falls below RandGuess, suggesting that zero-shot prompting and tuning-free LLMs do not yet fully capture sleep-related signals expressed in open-ended multimodal behavior and may need stronger prompting or more specialized modeling.

9 Study 4: LLM Prompting Strategies Comparison

Given prior works that advanced prompting can enhance LLM reasoning in mental health area [34, 63, 82], we next address **RQ4:** *"Do advanced prompting strategies further improve LLMs' performance beyond zero-shot inference?"*. To answer this question, we compare multiple prompting strategies, from zero-shot to retrieval-augmented and reasoning-based approaches, using the best-performing multimodal input from Section 6, all-video aggregation from Section 8, and GPT-5.2 [50] to isolate the effect of prompting.

Method. The most direct prompting technique is **zero-shot** [10], which we use for all prior LLM and VLM baselines in Sections 6 to 8. Building on this zero-shot setup (with a persona, task description, and step-by-step instructions), we add more advanced prompting features to test whether they further improve performance. Specifically, we compare the following four prompting strategies against GPT-TFA0 while using the same full multimodal, all-video aggregated input and GPT-5.2. **Chain-of-Thought (CoT)** [78] elicits an explicit reasoning trace before prediction (GPT-CoT-All). **Few-shot** [10] adds five fixed input-score examples from the regression training set (GPT-Fewshot-All). More advanced, **retrieval-augmented generation (RAG)** [35] dynamically retrieves the top-k most similar examples (k 5) for each participant from the regression training set, based on multimodal input representations (GPT-RAG-All). **RAG+CoT** combines retrieval and explicit reasoning (GPT-RAGCoT-All). Details of the few-shot and RAG implementations, along with example prompts for each prompting strategy, are provided in Sections E, F and F.3.

Results. Table 4 shows that advanced prompting strategies generally outperform zero-shot prompting, indicating that they help LLMs better interpret multimodal behavioral evidence, especially in SQS. Uunlike zero-shot settings (GPT-TFA0 and GPT-TFA0-All), where sleep quality remained difficult to predict, advanced prompting raises SQS accuracy well above RandGuess (0.20), from GPT-TFA0-All at 0.15 to 0.27 with CoT, 0.44 with fewshot, 0.42 with RAG, and 0.45 with RAG+CoT. This further confirms that sleep-related evidence is not absent from the multimodal input, but instead requires stronger guidance or better grounding for LLMs to extract it reliably. Compared with GPT-TFA0-All, CoT improves most surveys, with especially large gains on GAD-7 (+0.12), SQS (+0.12), and SWLS (+0.06), suggesting that explicit intermediate reasoning helps the model organize evidence more effectively. Both fewshot and RAG improve over zero-shot prompting, but in different ways: few-shot provides fixed examples, while RAG adaptively retrieves ones. Fewshot performs better on PSS (0.74), SQS (0.44), and UCLA (0.68), while RAG performs better on PHQ-9

Table 5: Matthews Correlation Coefficient (MCC) (mean ± std over three seeds) across all models in three studies. Best values per column are shown in bold. Advanced prompting and multi-response aggregation generally improve MCC over simpler baselines, with RAG+CoT performing strongest overall across most surveys.

	Method	GAD-7	MLQ	MSPSS	PHQ-9	PSS	SPANE	SQS	SWLS	UCLA
Regres.	Ridge	0.08 ± 0.00	0.15 ± 0.00	0.07 ± 0.00	0.00 ± 0.00	0.10 ± 0.00	0.07 ± 0.00	0.07 ± 0.00	0.06 ± 0.00	0.00 ± 0.00
	RandForest	0.05 ± 0.02	0.17 ± 0.04	0.14 ± 0.00	0.14 ± 0.04	0.00 ± 0.00	0.02 ± 0.00	0.16 ± 0.00	0.00 ± 0.02	0.08 ± 0.04
VLMs	LlaVA	0.03 ± 0.06	0.11 ± 0.17	0.10 ± 0.19	0.01 ± 0.03	0.02 ± 0.08	0.03 ± 0.06	0.10 ± 0.02	0.04 ± 0.10	0.03 ± 0.05
	Qwen	0.10 ± 0.05	0.21 ± 0.04	0.17 ± 0.08	0.10 ± 0.03	0.02 ± 0.04	0.17 ± 0.02	0.03 ± 0.05	0.04 ± 0.02	0.08 ± 0.04
LLMs	GPT-T0	0.09 ± 0.02	0.26 ± 0.03	0.33 ± 0.03	0.31 ± 0.02	0.02 ± 0.00	0.14 ± 0.02	0.01 ± 0.04	0.13 ± 0.03	0.13 ± 0.01
	GPT-TF0	0.13 ± 0.01	0.30 ± 0.01	0.36 ± 0.06	0.30 ± 0.01	0.02 ± 0.00	0.15 ± 0.04	0.05 ± 0.00	0.15 ± 0.02	0.14 ± 0.02
	GPT-TA0	0.09 ± 0.01	0.30 ± 0.02	0.35 ± 0.00	0.35 ± 0.01	0.02 ± 0.00	0.14 ± 0.02	0.07 ± 0.03	0.15 ± 0.01	0.14 ± 0.00
	DeepSeek	0.08 ± 0.13	0.21 ± 0.05	0.31 ± 0.15	0.31 ± 0.06	0.02 ± 0.08	0.16 ± 0.22	0.02 ± 0.07	0.17 ± 0.03	0.26 ± 0.05
	Llama	0.03 ± 0.01	0.26 ± 0.01	0.45 ± 0.00	0.25 ± 0.01	0.06 ± 0.03	0.22 ± 0.01	0.03 ± 0.00	0.19 ± 0.01	0.22 ± 0.02
	GPT-TFA0	0.09 ± 0.02	0.31 ± 0.03	0.37 ± 0.03	0.25 ± 0.02	0.02 ± 0.00	0.18 ± 0.01	0.05 ± 0.02	0.16 ± 0.02	0.15 ± 0.02
	GPT-TFA0-All	0.11 ± 0.03	**.39 ± . 1**	0.49 ± 0.03	0.27 ± 0.01	0.36 ± 0.03	0.33 ± 0.01	0.04 ± 0.03	0.18 ± 0.01	0.51 ± 0.01
	GPT-CoT-All	0.20 ± 0.03	0.25 ± 0.02	0.47 ± 0.02	0.24 ± 0.03	0.37 ± 0.04	0.37 ± 0.06	0.09 ± 0.05	0.27 ± 0.03	0.41 ± 0.02
	GPT-Fewshot-All	0.14 ± 0.02	0.33 ± 0.00	0.40 ± 0.00	0.34 ± 0.01	**.45 ± . 3**	0.37 ± 0.05	**.19 ± . 1**	0.17 ± 0.01	**.52 ± . 1**
	GPT-RAG-All	0.23 ± 0.02	0.32 ± 0.04	0.43 ± 0.01	**.36 ± . 1**	0.34 ± 0.07	0.36 ± 0.02	0.11 ± 0.04	0.22 ± 0.03	0.50 ± 0.01
	GPT-RAGCoT-All	**.27 ± . 2**	0.38 ± 0.01	**.52 ± . 1**	0.36 ± 0.06	0.42 ± 0.02	**.4 ± . 4**	0.18 ± 0.01	**.3 ± . 2**	0.45 ± 0.02

Table 6: Pearson Correlation (mean ± std over three seeds) across all models in three studies. Best values per column are shown in bold. Advanced prompting and multi-response aggregation generally improve Pearson correlation over simpler baselines, with all-video prompting and advanced strategies yielding the strongest performance on most surveys.

	Method	GAD-7	MLQ	MSPSS	PHQ-9	PSS	SPANE	SQS	SWLS	UCLA
Regres.	Ridge	0.10 ± 0.00	0.27 ± 0.00	0.18 ± 0.00	0.11 ± 0.00	0.07 ± 0.00	0.16 ± 0.00	0.06 ± 0.00	0.31 ± 0.00	0.03 ± 0.00
	RandForest	0.12 ± 0.01	0.23 ± 0.01	0.32 ± 0.01	0.33 ± 0.02	0.02 ± 0.03	0.11 ± 0.02	0.27 ± 0.02	0.26 ± 0.02	0.27 ± 0.01
VLMs	LlaVA	0.29 ± 0.02	0.23 ± 0.10	0.10 ± 0.17	0.03 ± 0.17	0.00 ± 0.10	0.13 ± 0.29	0.07 ± 0.05	0.21 ± 0.08	0.28 ± 0.11
	Qwen	0.18 ± 0.06	0.45 ± 0.08	0.46 ± 0.05	0.22 ± 0.00	0.25 ± 0.05	0.44 ± 0.02	0.06 ± 0.03	0.40 ± 0.06	0.31 ± 0.03
LLMs	GPT-T0	0.21 ± 0.02	0.47 ± 0.02	0.67 ± 0.02	0.58 ± 0.01	0.30 ± 0.01	0.44 ± 0.01	0.18 ± 0.12	0.61 ± 0.00	0.50 ± 0.01
	GPT-TF0	**.4 ± . 3**	0.48 ± 0.01	0.67 ± 0.01	0.61 ± 0.01	0.32 ± 0.01	0.43 ± 0.02	0.01 ± 0.07	0.61 ± 0.01	0.52 ± 0.01
	GPT-TA0	0.25 ± 0.01	0.47 ± 0.01	0.67 ± 0.00	0.54 ± 0.01	0.34 ± 0.00	0.44 ± 0.01	0.08 ± 0.08	0.62 ± 0.00	0.49 ± 0.01
	DeepSeek	0.36 ± 0.20	0.35 ± 0.10	0.48 ± 0.11	0.48 ± 0.12	0.14 ± 0.17	0.43 ± 0.23	0.43 ± 0.09	0.41 ± 0.03	0.57 ± 0.10
	Llama	0.26 ± 0.02	0.48 ± 0.01	0.59 ± 0.01	0.54 ± 0.01	0.31 ± 0.03	0.30 ± 0.00	0.52 ± 0.00	0.51 ± 0.03	0.45 ± 0.00
	GPT-TFA0	0.17 ± 0.02	0.48 ± 0.01	0.67 ± 0.01	0.58 ± 0.04	0.36 ± 0.02	0.42 ± 0.02	0.02 ± 0.06	0.62 ± 0.01	0.52 ± 0.01
	GPT-TFA0-All	0.13 ± 0.04	0.46 ± 0.01	**.74 ± . 3**	**.68 ± . 1**	**.67 ± . 1**	0.67 ± 0.01	0.01 ± 0.15	0.68 ± 0.00	0.72 ± 0.00
	GPT-CoT-All	0.31 ± 0.05	0.38 ± 0.02	0.62 ± 0.04	0.56 ± 0.07	0.61 ± 0.01	**.7 ± . 2**	0.46 ± 0.13	0.66 ± 0.03	0.65 ± 0.03
	GPT-Fewshot-All	0.35 ± 0.01	0.49 ± 0.02	0.72 ± 0.01	0.63 ± 0.01	0.64 ± 0.01	0.70 ± 0.01	0.51 ± 0.00	**.7 ± .**	**.73 ± . 1**
	GPT-RAG-All	0.24 ± 0.05	0.49 ± 0.01	0.73 ± 0.01	0.65 ± 0.00	0.65 ± 0.01	0.65 ± 0.02	0.50 ± 0.03	0.70 ± 0.00	0.70 ± 0.01
	GPT-RAGCoT-All	0.34 ± 0.08	**.5 ± . 4**	0.65 ± 0.02	0.63 ± 0.03	0.61 ± 0.03	0.65 ± 0.02	**.52 ± . 1**	0.66 ± 0.02	0.67 ± 0.02

(0.61) and remains competitive on MLQ (0.64), MSPSS (0.80), and SWLS (0.39). One possible explanation is that when retrieval fails to find sufficiently similar examples, RAG may provide weaker guidance than few-shot prompting. Combining retrieval and reasoning, RAG+CoT performs best overall, achieving the highest or tied-highest accuracy on most of the surveys, including GAD-7 (0.54), MLQ (0.67), MSPSS (0.82), SPANE (0.62), SQS (0.45), and SWLS (0.45), while tying the best results on PHQ-9 and PSS. Notably, CoT alone (0.61) lowers performance on UCLA compared with GPT-TFA0-All (0.67), suggesting that explicit reasoning may become overly conservative or miscalibrated for loneliness-related judgments. Although RAG+CoT (0.63) does not improve UCLA over RAG alone (0.67), it performs better than CoT alone, indicating that retrieval helps stabilize reasoning.

10 Discussion

Our study examines whether LLMs can assess psychological well-being from naturalistic open-ended video responses. Across four studies, we found that success depends not just on adding more modalities but on effective representation, aggregation, and reasoning over behavioral evidence. Specifically, the strongest predictive performance arises when the model is given structured transcript, acoustic, and facial information that enable coherent psychometric judgment rather than relying solely on raw multimodal access.

First, we found that **multimodal input can enhance LLMs' predictive performance in well-being assessment, especially when meaningful cues are conveyed through nonverbal and paraverbal features, rather than explicit linguistic content** (RQ1; Table 1). Transcript-only input performs strongly for measures, such as MLQ (meaning in life), MSPSS (perceived social support), and SWLS (life satisfaction), where participants can directly articulate their beliefs, relationships, and life evaluations in language. In contrast, full multimodal input yields greater benefits for GAD-7 (anxiety), PHQ-9 (depression), PSS (stress), SPANE (affective well-being), and UCLA (loneliness), where signals are reflected not only in what is said but also in how it is expressed. These patterns suggest that multimodal cues are especially valuable when well-being states are conveyed more indirectly through tone, prosody, facial expressions, and behavior, rather than explicitly verbalized. Importantly, the advantage of multimodal input is not merely in adding more data, but in providing complementary evidence about individuals' distress and social-emotional states that may not be fully captured through words alone.

Second, we found that **LLMs outperform both traditional regression baseline models and VLMs overall** (RQ2; Table 2). Regression baselines remain competitive on selected surveys, especially SQS (Table 2), which suggests that untuned LLMs with simple prompting may miss some signals that trained regression models can capture. VLMs performed worse than the strongest LLM-based models, despite their ability to process image/video frames. This suggests that current state-of-the-art VLMs are not yet well-suited for nuanced psychological assessment using raw visual signals alone. Instead, the strongest approach appears to involve translating multimodal data into psychologically meaningful, text-compatible summaries that leverage the advanced reasoning capabilities of LLMs.

Third, **providing additional contextual information from other well-being dimensions consistently improves LLMs' predictive performance for a specific well-being construct** (RQ3; Table 3). This suggests that the model benefits from repeated and stable behavioral evidence across a wider range of contexts. In addition, **advanced prompting strategies unlock deeper LLM reasoning capabilities** (RQ4; Table 4). A critical example is the SQS: while zero-shot attempts failed to surpass random guessing and regression baselines, advanced prompting successfully extracted the latent sleep-related signals, elevating performance significantly. Overall, RAG+CoT performs best among the prompting techniques. However, advanced prompting are not always beneficial: for example, CoT alone degrades performance on UCLA, suggesting that explicit reasoning may over-analyze or misinterpret the subtle, sometimes contradictory social cues associated with perceived loneliness. Future work should therefore examine the models more deeply by identifying the specific behavioral signals that drive predictions and by examining more closely when prompting affects performance, thereby improving interpretability beyond overall performance.

An important limitation of our study lies in the underlying score distribution of the dataset. As a preliminary investigation based on a predominantly college-aged, non-clinical sample, several distress-oriented scales are strongly skewed toward lower-severity ranges. For example, 57% of participants fall into the minimal anxiety level in GAD-7. Full ground-truth score statistics are reported in Table 8, and severity (Figure 2) and raw score distributions (Figure 3) for all surveys are provided in Section A in the Supplementary Material. This skew can make raw accuracy misleading, as models may appear strong by favoring dominant lower-severity classes, and it also limits generalizability to broader or more clinically acute populations.

To better account for this imbalance, we additionally report the Matthews Correlation Coefficients (MCC) [42], a metric that is more robust under class imbalance [17], across all models and studies (Table 5). The MCC results reinforce the same overall conclusion. Advanced prompting and all-video aggregation do not merely improve average prediction metrics, but also improve the model's ability to distinguish less frequent, more acute well-being levels under skewed label distributions. For example, although Ridge achieves higher category accuracy than GPT-TFA0 on GAD-7 (0.40 vs. 0.37), its MCC is lower (0.08 vs. 0.09), suggesting weaker discrimination once imbalance is taken into account. Across all investigated methods, RAG+CoT yields the strongest MCC overall across most surveys, while few-shot performs especially well on PSS, SQS, and UCLA.

Beyond categorical accuracy, Pearson correlation analyses (Table 6) showed that LLM-based methods also better align with the underlying survey scores. This shows that the models do not merely place users into the correct severity buckets, but also capture the relative intensity and nuanced variation of their well-being states. Taken together, the correlation analyses results show that our multimodal LLM framework not only maintains strong alignment with the continuous structure of psychological well-being but also remains more robust under real-world dataset imbalance.

11 Conclusion

We investigate whether LLMs can predict psychological well-being from naturalistic open-ended video responses. We find that strong performance depends not simply on adding more modalities, but on how multimodal behavioral evidence is represented, aggregated, and reasoned over. Structured multimodal inputs that combine transcripts with acoustic and facial cues improve sensitivity to psychological distress. Performance further improves when evidence is aggregated across all video responses rather than drawn from a single survey-specific response. Beyond input structure, advanced prompting yields additional boost. Our results suggest that retrieve-then-reason prompting produces the most robust and well-calibrated predictions by grounding inference in relevant reference cases, especially for subtle or weakly expressed survey signals. Together, these findings suggest that LLM-based conversational systems can potentially serve as scalable, ecologically valid complements to existing monitoring frameworks by enabling more frequent, low-burden assessment between clinical visits and supporting more proactive, personalized mental health care.

Safe and Responsible Innovation Statement

As AI becomes more integrated into mental health assessment, our work explores LLMs to predict psychological well-being from naturalistic, open-ended video responses. This framework is intended for low-burden longitudinal assessment, not diagnosis, triage, or autonomous decision-making, and requires informed consent, strong privacy protections, and secure handling of multimodal data. Given our predominantly college-aged, non-clinical sample, we acknowledge risks of bias, misclassification, and overgeneralization; thus, we position this work as a step toward human-centered support systems that complement, rather than replace, validated psychometric tools and clinical judgment.

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